

E-Commerce SQL Analysis Project

Analyze e-commerce order data to understand revenue performance, customer value, payment behavior, and operational issues such as cancellations and returns.

Monthly Revenue Analysis

02

SQL QUERY

```
SELECT FORMAT(order_date, 'yyyy-MM') AS YearMonth,  
       SUM(quantity * price)          AS Revenue  
FROM   ecommerce_orders  
WHERE  order_status = 'Delivered'  
GROUP BY FORMAT(order_date, 'yyyy-MM')  
ORDER BY YearMonth;
```

YearMonth	Revenue
2024-01	450287.47
2024-02	411865.57
2024-03	467168.77
2024-04	418844.68
2024-05	486079.49
2024-06	423814.79
2024-07	472779.67
2024-08	463496.41
2024-09	451780.72
2024-10	425292.34
2024-11	461920.59
2024-12	408755.11

Insight

Identifies seasonal revenue patterns and monthly sales performance to guide inventory and marketing decisions.

Top 5 Categories by Revenue

03

SQL QUERY

```
SELECT TOP 5
    category,
    ROUND(SUM(quantity * price),2) AS Revenue
FROM    ecommerce_orders
WHERE   order_status = 'Delivered'
GROUP BY category
ORDER BY Revenue DESC;
```

Category	Revenue
Beauty	946625.37
Sports	914307.38
Books	885037.54
Electronics	884587.62
Fashion	864491.09

Insight

Shows which product segments drive the most revenue — critical for promotional budgeting and shelf prioritization.

Cancellation Rate by City

04

SQL QUERY

```
SELECT city,  
       SUM(CASE WHEN order_status = 'Cancelled'  
               THEN 1 ELSE 0 END) * 1.0  
       / COUNT(order_id) AS cancellation_rate  
FROM   ecommerce_orders  
GROUP BY city  
ORDER BY cancellation_rate DESC;
```

city	Cancellation_rate
Ahmedabad	9.195402298800
Delhi	8.988764044900
Pune	8.221225710000
Kolkata	8.133971291800
Hyderabad	8.052708638300
Mumbai	8.038585209000
Chennai	7.368421052600
Bangalore	6.867671691700

Insight

Cities with high cancellation rates may signal logistics, delivery SLA, or customer-service issues requiring intervention.

SQL QUERY

```
SELECT
  (SUM(CASE WHEN payment_method='COD' THEN 1 ELSE 0 END)*1.0/COUNT(order_id))*100 AS COD_Pct,
  (SUM(CASE WHEN payment_method='UPI' THEN 1 ELSE 0 END)*1.0/COUNT(order_id))*100 AS UPI_Pct,
  (SUM(CASE WHEN payment_method='Card' THEN 1 ELSE 0 END)*1.0/COUNT(order_id))*100 AS Card_Pct
FROM ecommerce_orders;
```

COD_PCT	UPI_PCT	COD_PCT
20.52	49.86	29.62

Insight

Reveals customer payment preferences, informing checkout UX improvements and payment gateway investment decisions.

Top 10 Customers by Spending

05

SQL QUERY

```
SELECT TOP 10
    customer_id,
    ROUND(SUM(quantity * price),2) AS Revenue
FROM    ecommerce_orders
WHERE   order_status = 'Delivered'
GROUP BY customer_id
ORDER BY Revenue DESC;
```

customer_id	Revenue
1669	21571.53
1112	20819.74
1098	19176.25
1786	18440.84
1525	17875.21
1116	17268.74
1647	17089.63
1914	15575.44
1038	15495.31
1658	15308.38

Insight

Pinpoints high-value customers for loyalty programs, VIP offers, and churn-prevention strategies.

Top 10 Products by Quantity Sold

07

SQL QUERY

```
SELECT TOP 10
    product_id,
    SUM(quantity) AS Units_Sold
FROM    ecommerce_orders
WHERE   order_status = 'Delivered'
GROUP BY product_id
ORDER BY Units_Sold DESC;
```

product_id	Unit_Sold
146	87
17	87
166	84
188	81
122	81
112	80
183	80
187	77
195	76
164	75

Insight

Highlights best-selling products for demand forecasting, inventory replenishment, and supplier negotiations.

Return Rate by Category

08

SQL QUERY

```
SELECT category,  
       ROUND((SUM(CASE WHEN order_status = 'Returned'  
                       THEN 1 ELSE 0 END) * 1.0  
              / COUNT(*))*100,2) AS Return_Rate  
FROM   ecommerce_orders  
GROUP BY category  
ORDER BY Return_Rate DESC;
```

category	Return_Rate
Sports	7.98
Fashion	7.45
Home	7.04
Electronics	6.77
Books	6.11
Beauty	6.06

Insight

Categories with elevated return rates may indicate product quality gaps, misleading descriptions, or sizing issues.

Month-over-Month Revenue Growth

09

SQL QUERY

```
WITH month_revenue AS (  
    SELECT FORMAT(order_date,'yyyy-MM') AS  
    YearMonth, SUM(quantity*price) AS Revenue  
    FROM ecommerce_orders  
    WHERE order_status='Delivered'  
    GROUP BY FORMAT(order_date,'yyyy-MM')  
)  
SELECT YearMonth, Revenue,  
LAG(Revenue) OVER(ORDER BY YearMonth) AS  
Prev_Revenue,  
ROUND((Revenue - LAG(Revenue) OVER(ORDER BY YearMonth))/  
NULLIF(LAG(Revenue) OVER(ORDER BY YearMonth)*100,0),2)  
AS MoM_PCT  
FROM month_revenue;
```

YearMonth	Revenue	prev_revenue	MoM_PCT
2024-01	450287.47	NULL	NULL
2024-02	411865.57	450287.47	-8.53
2024-03	467168.77	411865.57	13.43
2024-04	418844.68	467168.77	-10.34
2024-05	486079.49	418844.68	16.05
2024-06	423814.79	486079.49	-12.81
2024-07	472779.67	423814.79	11.55
2024-08	463496.41	472779.67	-1.96
2024-09	451780.72	463496.41	-2.53
2024-10	425292.34	451780.72	-5.86
2024-11	461920.59	425292.34	8.61
2024-12	408755.11	461920.59	-11.51

Insight

Uses window functions (LAG) to compute month-on-month growth, enabling trend analysis and forecasting.

Key Business Insights



Revenue Trends

Monthly analysis reveals seasonal demand peaks — valuable for campaign timing.



Category Performance

A small number of categories account for the majority of delivered revenue.



High-Value Customers

Top 10 customers disproportionately drive sales — ideal targets for loyalty perks.



Payment Preferences

Payment mix insights improve checkout strategy and gateway investment decisions.



Operational Improvements

Cancellation & return analysis pinpoints cities and categories needing intervention.